

Near-Earth Object Detection - ML

Machine Learning Final Project
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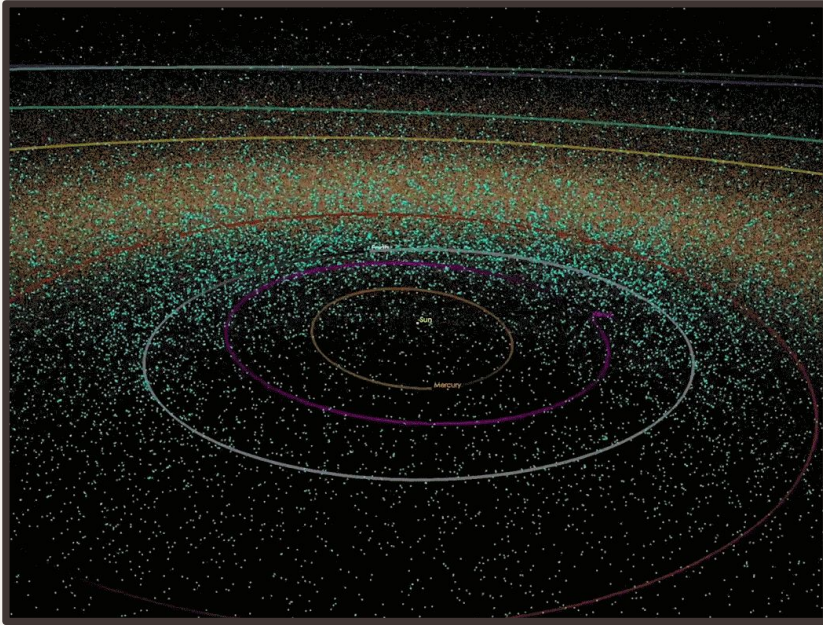
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1. Research Goal



Leverage machine learning techniques to detect dangerous Near-Earth Objects (NEOs).

2. Importance

**NASA TEAMS WEIGH
NUCLEAR RESPONSE AS
ASTEROID 2024 YR4
HOLDS 4% CHANCE OF
HITTING THE MOON**



Nina Singh-Hudson

November 27, 2025

The Problem: NEOs can pose a potential, significant threat.

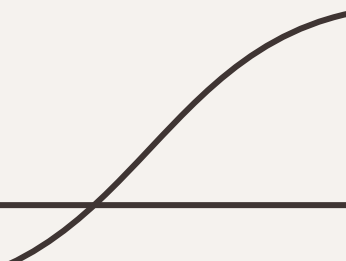
Solution Importance:
Detection of dangerous space objects can allow early action to prevent catastrophe.

3. Dataset

Data is composed of 24,000 detected objects from the NASA near earth object database.

One issue in our dataset is that only **9% are labeled as dangerous.**

The dataset includes columns such as:

- Absolute Magnitude
 - Estimated Diameter
 - Orbit Class Type
 - Perihelion Distance
 - Aphelion Distance
 - Danger to Earth
- 

4. SOTA/Literature Review

The SOTA model is from Ved Umrajkar. The weaknesses we saw in the SOTA are:

- No Feature Engineering
- No Hyperparameter Tuning
- Data Leakage

SOTA

The SOTA accuracy is as follows:

Model:	Accuracy (%):
Decision Tree	90.7
Random Forest	91.9
Logistic Regression	91.8
LDA	90.5

Contributions

Our contribution to the existing field of work is the **refinement of existing models** and the addition of **neural network**.

```
# Build model
nn_model = MLPClassifier(
    hidden_layer_sizes=(64, 32),
    activation='relu',
    solver='adam',
    max_iter=500,
    random_state=42
)
```


Feature Engineering

The SOTA model did not include any feature engineering. Random Forest had the highest accuracy so we focused on interaction features.

We added the parameters:

- Mean Diameter
- Observation Span
- Semi Major Axis
- Eccentricity
- Orbit Radius
- Inverse Perihelion
- Diameter * Eccentricity
- Magnitude * Diameter

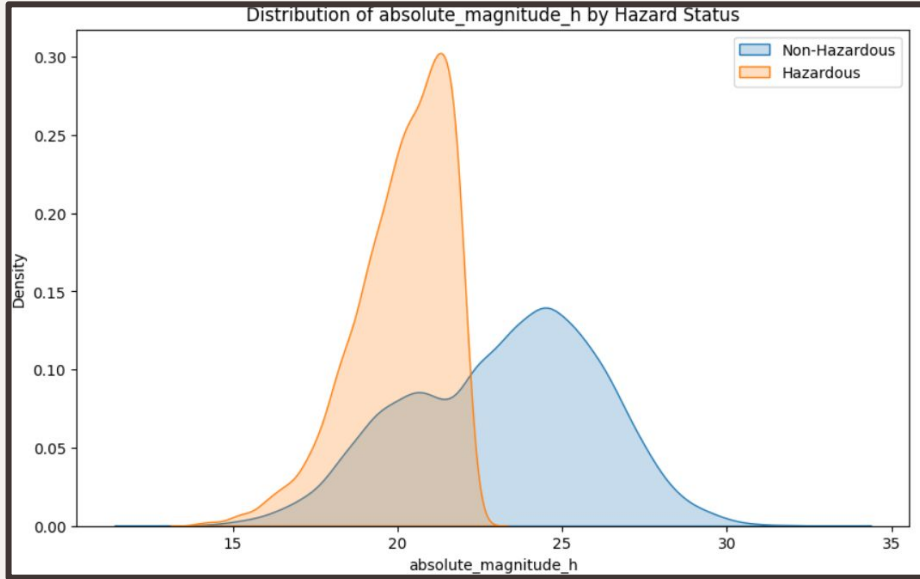
Feature Engineering

The dataset lacked data entries in 16 rows. The SOTA used the global mean. We choose to remove these rows.

The columns lacking data are:

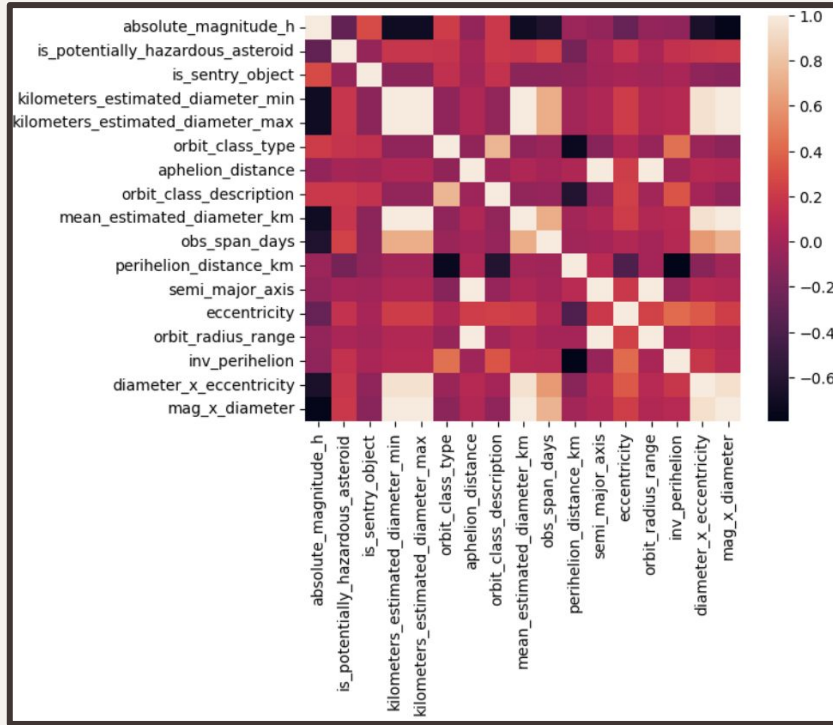
- Diameter max
- Diameter min
- Absolute magnitude
- Last observation date
- First observation date

Feature Engineering



Hazardous and Non-Hazardous samples were divided and parameters were evaluated.

Feature Engineering



is_potentially_hazardous_asteroid	
absolute_magnitude_h	-0.298140
is_potentially_hazardous_asteroid	1.000000
is_sentry_object	-0.058802
kilometers_estimated_diameter_min	0.175068
kilometers_estimated_diameter_max	0.175068
orbit_class_type	0.167707
aphelion_distance	0.009924
orbit_class_description	0.201452
mean_estimated_diameter_km	0.175068
obs_span_days	0.245532
perihelion_distance_km	-0.206436
semi_major_axis	-0.000796
eccentricity	0.158862
orbit_radius_range	0.020676
inv_perihelion	0.150913
diameter_x_eccentricity	0.176738
mag_x_diameter	0.199024

Hyper Parameter Tuning

For the decision tree, random forest, and logistic regression models, we used a random search algorithm. For LDA we decided a random search would be too unstable, so we used a grid search.

Model:	Decision Tree
Depth:	7
Sample Leafs:	9
Sample Split:	9

Model:	Random Forest
Depth:	9
Sample Leafs:	2
Sample Split:	4
features:	0.5

Model:	Logistic Regression
C:	7.32
Penalty:	L2
Solver:	Saga

Model:	LDA
Shrinkage:	0.1
Solver:	lsqr

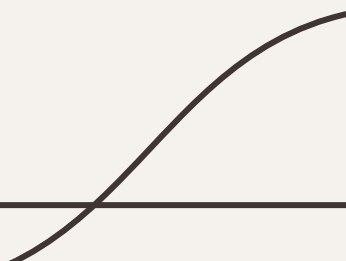
Additional Improvements

While SOTA methods include the raw **detection date**, we determined it to be ineffective because the observation timestamp is not physically related to hazard risk and behaves as a noisy, non-informative feature. We replaced this data with the **observation duration**.

The SOTA Includes the parameter “**sentry object**”, these are potentially dangerous objects already tracked by NASA. We removed this parameter as it would be ineffective if the model was given a newly discovered near earth object.

Final Optimized Models

Our final models Included the parameters:

- Orbit Class Description
 - Perihelion Distance
 - Observation Span
 - Absolute magnitude
 - Diameter * Eccentricity
 - Magnitude * Diameter
 - Eccentricity
 - Orbit Radius Range
 - Mean Diameter
- 

5. Results

The final models produced the results below:

Model:	Accuracy (%):
Decision Tree	91.7
Random Forest	92.5
Logistic Regression	91.7
LDA	91.4

Neural Network

We tested an additional model, an MLP classifier. The parameters were:

Layers:	2
Neurons:	64, 32
Activation:	ReLU

Neural Network

The NN model produced the results below:

Pros:	Cons:
Learns nonlinear decision boundaries automatically.	Requires scaling and careful parameterization
Model interactions between features more effectively than linear models.	
Scales well with more features and data	

```
Neural Network Accuracy: 0.9212007504690432

Classification Report:
              precision    recall  f1-score   support

     0           0.94       0.97       0.96       4378
     1           0.57       0.38       0.46        419

   accuracy              0.92       4797
  macro avg           0.76       0.68       0.71       4797
 weighted avg           0.91       0.92       0.91       4797

Confusion Matrix:
[[4261  117]
 [ 261 158]]
```

Model Comparison/Evaluation

Random Forest performs better than other models

- It handles nonlinear features better
- Less sensitive to overfitting

Improvement from SOTA: 91.9% to 92.5%

- Benefits from Feature Engineering in detecting non-hazardous vs hazardous objects

6. Conclusion

- Classification Problem
 - Data + Features
 - Model Comparison(LR, RF, NN)
 - Potential Work
-

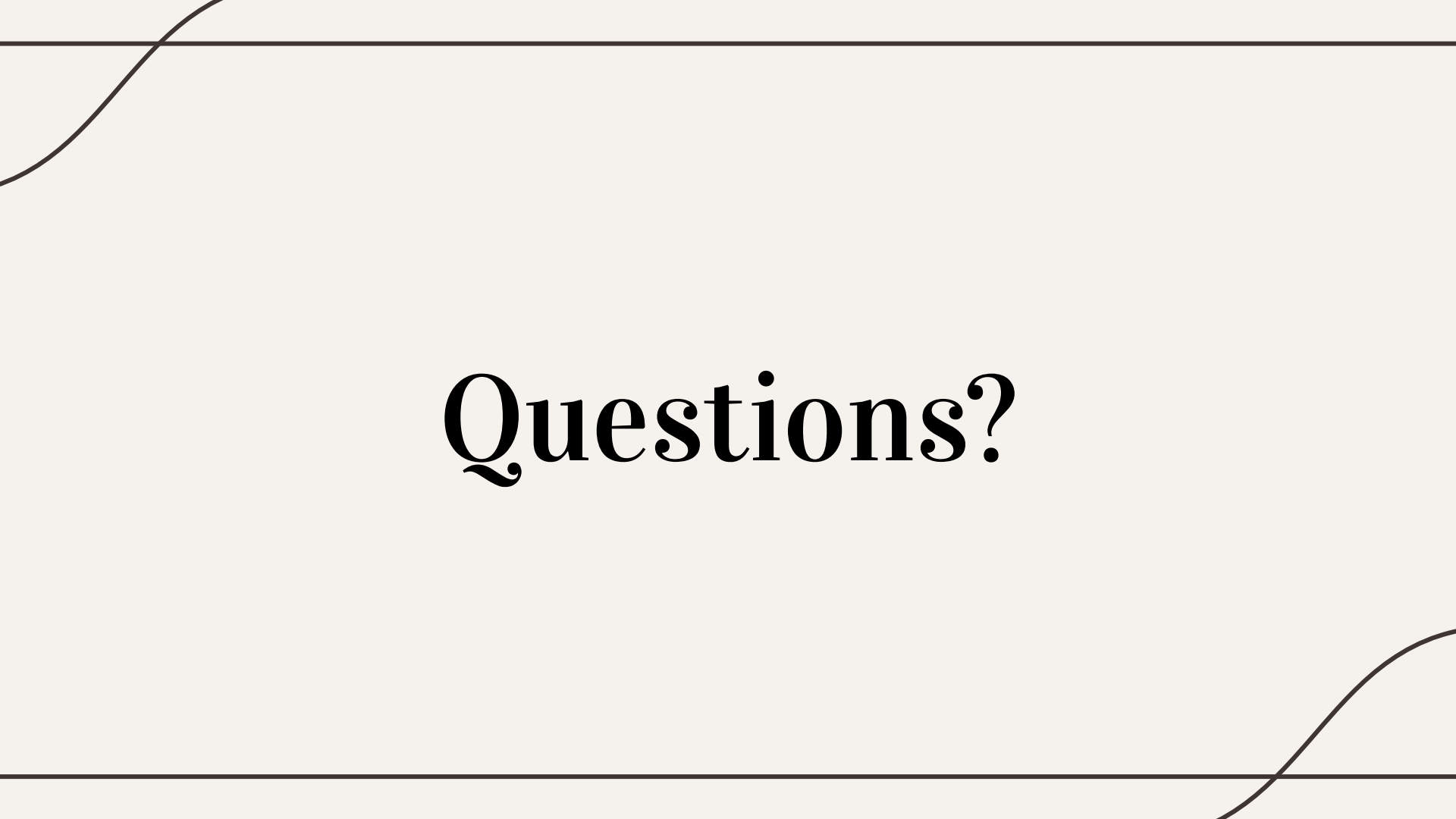
Outlook

Future improvements to the model include:

- Using an larger, expanded dataset
- Additional “dangerous” samples
- Using more advanced NN models
- Incorporating Sequence Modeling, like LSTMs

References

- “Twenty Years of Tracking Near-Earth Objects.” NASA, NASA, www.jpl.nasa.gov/news/twenty-years-of-tracking-near-earth-objects/. Accessed 1 Dec. 2025.
- Ramachandran, Aditya. “NASA Near Earth Objects Information.” Kaggle, 9 July 2022, www.kaggle.com/datasets/adityaramachandran27/nasa-near-earth-objects-information/data.
- NASA Jet Propulsion Lab: <https://cneos.jpl.nasa.gov/ca/>

The image features two thin, dark horizontal lines that span the width of the frame. Each line has a smooth, curved end on the left and right sides, creating a subtle, elegant border.

Questions?